How to implement a DAS Listener in C#

Reference : RA 468

Welcome to the documentation for the **RA_0468 DAS Listener**.
This reference architecture demonstrates how to build a responsive Data Analytics Solution (DAS) that listens for messages from a test program and processes them dynamically.

General Technical Info

Component	**Version**
Language	C#
Framework	.NET 8
Environment	Windows 10
IGXL	> 10.30.10
MST	> 2016C
IDE	Visual Studio Code / Visual Studio 2022

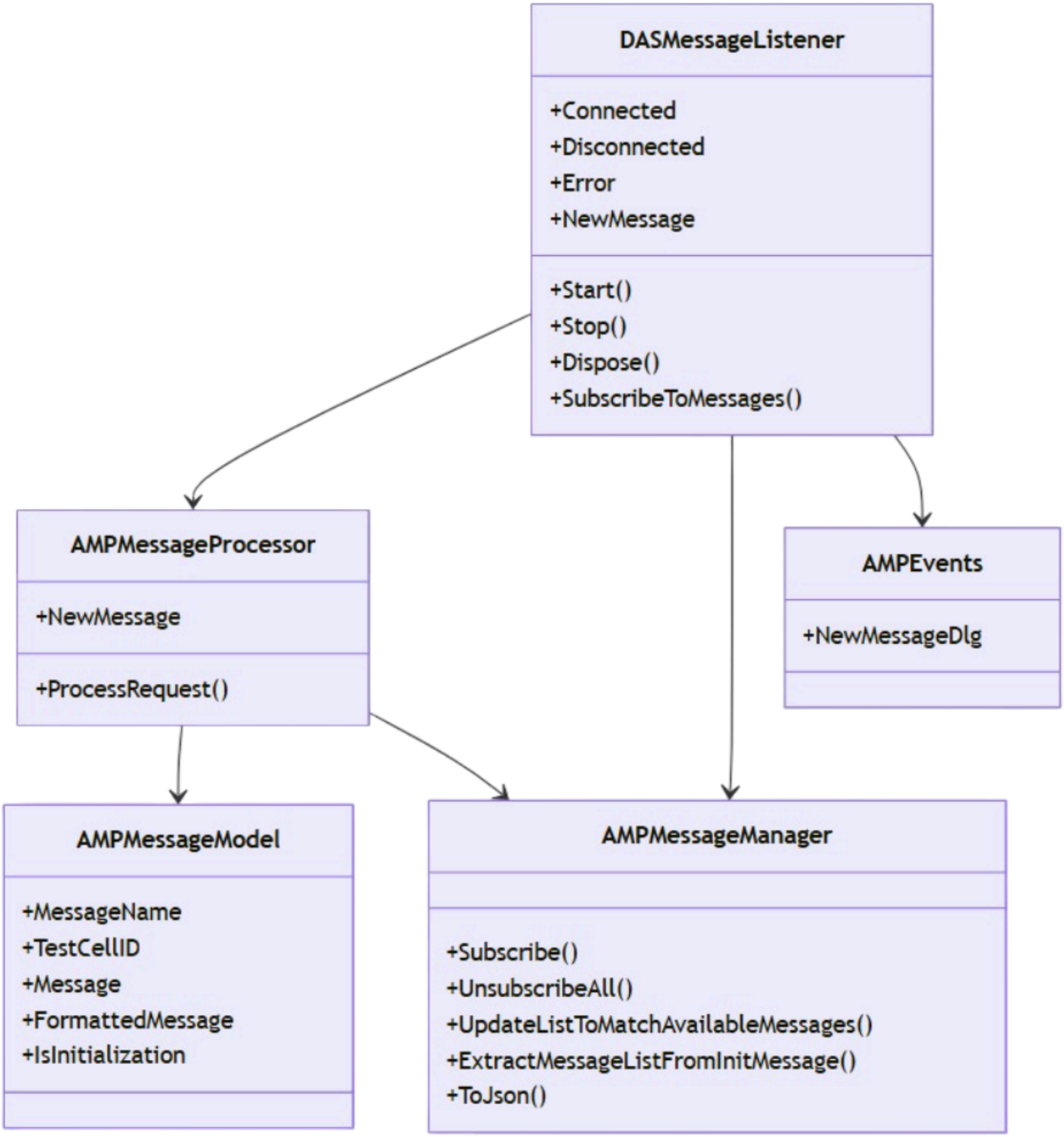
Key Features

- Real-time HTTP message reception using `HttpListener`
- Events for connection, disconnection, errors, and new message dispatch
- Structured message model including timestamps, URL segments, and raw/parsed payload
- Fully documented C# code using .NET 8

Architecture Overview

The DAS acts as a lightweight server that receives messages from a test cell. Each message is parsed, validated, and dispatched to an event system.

Project Structure Diagram



Example: Getting Started

Here's a quick example of how to instantiate and use the `DASMessageListener` class:

```csharp
using Teradyne.Archimedes.AMPCommunication;
using Teradyne.Archimedes.DAS.Listener;

public class Program
{
    public static async Task Main(string[] args)
    {
        Console.WriteLine("Hello, Welcome to my DAS!");

        var listener = new DASMessageListener("http://localhost:3000/tems/");

        listener.Connected += () => Console.WriteLine("Connected to DAS.");
        listener.Disconnected += () => Console.WriteLine("Disconnected.");
        listener.Error += (ex) => Console.WriteLine($"Error: {ex.Message}");
        listener.NewMessage += (cellid, message) =>
        {
            Console.WriteLine($"New message from {cellid}: {message.MessageName}");
        };

        await listener.Start();
    }
}
```

This documentation is part of the Teradyne Archimedes Reference Architecture series.

Namespace Teradyne.Archimedes.AMPCommunication

Classes

AMPEvents

Provides utility definitions for the DAS system, including event delegate declarations.

Delegates

AMPEvents.NewMessageDlg

Delegate for handling new message events. Triggered when a new AMPMessageModel is received from a test cell.

Class AMPEvents

Namespace: Teradyne.Archimedes.AMPCommunication

Assembly: Teradyne.Archimedes.DAS.Listener.dll

Provides utility definitions for the DAS system, including event delegate declarations.

```
public class AMPEvents
```

Inheritance

object ← AMPEvents

Inherited Members

object.Equals(object), object.Equals(object, object), object.GetHashCode(), object.GetType(), object.MemberwiseClone(), object.ReferenceEquals(object, object), object.ToString()

Delegate AMPEvents.NewMessageDlg

Namespace: Teradyne.Archimedes.AMPCommunication

Assembly: Teradyne.Archimedes.DAS.Listener.dll

Delegate for handling new message events. Triggered when a new AMPMessageModel is received from a test cell.

```
public delegate void AMPEvents.NewMessageDlg(string cellid, AMPMessageModel message)
```

Parameters

`cellid` string

The identifier of the test cell that received the message.

`message` AMPMessageModel

The AMP message payload.

Namespace Teradyne.Archimedes.AMPCommunication.Messages

Classes

AMPMessageManager

Manages AMP message subscriptions and interactions. It is mandatory to return a list of message names as a response to the INITIALIZATION message and recommended for each message.

Class AMPMessageManager

Namespace: Teradyne.Archimedes.AMPCommunication.Messages

Assembly: Teradyne.Archimedes.DAS.Listener.dll

Manages AMP message subscriptions and interactions. It is mandatory to return a list of message names as a response to the INITIALIZATION message and recommended for each message.

```
public class AMPMessageManager
```

Inheritance

object ← AMPMessageManager

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

MessageList

Current list of subscribed messages.

```
public HashSet<string> MessageList { get; }
```

Property Value

HashSet<string>

Methods

ExtractMessageListFromInitMessage(AMPMessageModel)

Extracts the message list from an INITIALIZATION message.

```
public void ExtractMessageListFromInitMessage(AMPMessageModel message)
```

Parameters

`message` AMPMessageModel

The INITIALIZATION message model.

Subscribe(IEnumerable<string>)

Subscribes to multiple message names.

```
public void Subscribe(IEnumerable<string> messageNames)
```

Parameters

`messageNames` IEnumerable<string>

A collection of message names to subscribe to.

Subscribe(string)

Subscribes to a single message name if it is valid.

```
public bool Subscribe(string messageName)
```

Parameters

`messageName` string

The name of the message to subscribe to.

Returns

bool

True if subscribed, false if already present or invalid.

SubscribeAll()

Subscribes to all predefined AMP messages.

```
public void SubscribeAll()
```

ToJson()

Serializes the message list into a JSON string.

```
public string ToJson()
```

Returns

string

A JSON string containing the MESSAGE_LIST.

UnsubscribeAll()

Clears the current message subscription list.

```
public void UnsubscribeAll()
```

UpdateListToMatchAvailableMessages(AMPMessageManager)

Intersects the message list with another manager's list. Only shared messages are kept.

```
public void UpdateListToMatchAvailableMessages(AMPMessageManager available)
```

Parameters

`available` AMPMessageManager

Manager with the available message list.

Namespace Teradyne.Archimedes.AMPCommunication.Models

Classes

AMPMessageModel

Represents an AMP message with metadata, test cell ID, timestamps, and JSON message formatting.

Class AMPMessageModel

Namespace: Teradyne.Archimedes.AMPCommunication.Models

Assembly: Teradyne.Archimedes.DAS.Listener.dll

Represents an AMP message with metadata, test cell ID, timestamps, and JSON message formatting.

```
public class AMPMessageModel
```

Inheritance

object ← AMPMessageModel

Inherited Members

object.Equals(object), object.Equals(object, object), object.GetHashCode(), object.GetType(), object.MemberwiseClone(), object.ReferenceEquals(object, object), object.ToString()

Properties

FormattedMessage

The JSON-formatted version of the message. Falls back to raw message if invalid.

```
public string FormattedMessage { get; }
```

Property Value

string

IsInitialization

Determines whether the message name indicates an initialization command.

```
public bool IsInitialization { get; }
```

Property Value

bool

Message

The raw JSON message string. When set, attempts to format it with indentation.

```
public string Message { get; set; }
```

Property Value

string

MessageName

The name of the message extracted from the URL.

```
public string MessageName { get; set; }
```

Property Value

string

TestCellID

The identifier of the test cell associated with the message.

```
public string TestCellID { get; set; }
```

Property Value

string

TimeStamp

The timestamp of the message, either parsed or defaulting to UTC now.

```
public DateTime TimeStamp { get; set; }
```

Property Value

[DateTime]

URL

The URL segments associated with the message.

```
public string[] URL { get; set; }
```

Property Value

[string][]

Namespace Teradyne.Archimedes.DAS.Listener

Classes

DASMessageListener

DASMessageListener is responsible for starting and stopping the DAS HTTP listener, receiving incoming HTTP requests, and dispatching them to a processor. It manages event notifications for connection, disconnection, errors, and new messages.

Delegates

DASMessageListener.DasErrorHandler

Delegate when the DAS is facing to an error

DASMessageListener.DasEventHandler

Delegate used for the event related to the DAS

Class DASMessageListener

Namespace: Teradyne.Archimedes.DAS.Listener

Assembly: Teradyne.Archimedes.DAS.Listener.dll

DASMessageListener is responsible for starting and stopping the DAS HTTP listener, receiving incoming HTTP requests, and dispatching them to a processor. It manages event notifications for connection, disconnection, errors, and new messages.

```
public class DASMessageListener : IDisposable
```

Inheritance

object ← DASMessageListener

Implements

IDisposable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DASMessageListener(string)

Initializes a new instance of the DASMessageListener class with a DAS URL.

```
public DASMessageListener(string dasurl)
```

Parameters

`dasurl` string

The URL to listen to.

Exceptions

ArgumentNullException

Fields

_messageList

System for managing the list of messages we want to receive

```
protected AMPMessageManager _messageList
```

Field Value

AMPMessageManager

Properties

DASURL

Gets the DAS listener URL.

```
public string DASURL { get; }
```

Property Value

string

Methods

Dispose()

Disposes the listener and stops it if necessary.

```
public void Dispose()
```

Start()

Starts the HTTP listener and begins processing incoming requests.

```
public Task<bool> Start()
```

Returns

Task<bool>

True if started successfully, false otherwise.

Stop()

Stops the HTTP listener and cancels all ongoing tasks.

```
public void Stop()
```

SubscribeToMessages(List<string>, bool)

Subscribes to a set of message names, optionally resetting previous subscriptions.

```
public void SubscribeToMessages(List<string> messageNames, bool resetFirst = true)
```

Parameters

`messageNames` List<string>

List of message names to subscribe to.

`resetFirst` bool

Whether to clear existing subscriptions first.

Events

Connected

Event when the DAS is connected

```
public event DASMessageListener.DasEventHandler Connected
```

Event Type

DASMessageListener.DasEventHandler

Disconnected

Event when the DAS is disconnected

```
public event DASMessageListener.DasEventHandler Disconnected
```

Event Type

DASMessageListener.DasEventHandler

Error

Event when the DAS encouterred and error

```
public event DASMessageListener.DasErrorHandler Error
```

Event Type

DASMessageListener.DasErrorHandler

NewMessage

Event when the DAS received a new message

```
public event AMPEvents.NewMessageDlg NewMessage
```

Event Type

AMPEvents.NewMessageDlg

Delegate DASMessageListener.DasErrorHandler

Namespace: Teradyne.Archimedes.DAS.Listener

Assembly: Teradyne.Archimedes.DAS.Listener.dll

Delegate when the DAS is facing to an error

```
public delegate void DASMessageListener.DasErrorHandler(Exception exception)
```

Parameters

`exception` Exception

Delegate DASMessageListener.DasEventHandler

Namespace: Teradyne.Archimedes.DAS.Listener

Assembly: Teradyne.Archimedes.DAS.Listener.dll

Delegate used for the event related to the DAS

```
public delegate void DASMessageListener.DasEventHandler()
```

📄 Description	🔗 Download Link
Entire Solution	The solution
GIT Hub Link	Git Hub

More details about those source code here

This documentation is part of the Teradyne Archimedes Reference Architecture series.